EEE 499B

Senior Design Project (Capstone) Report

NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

NAME OF PROJECT

MULTI-TASKING AUTONOMOUS DEVICE

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Letter of Transmittal

25 June 2023

To
Dr. Rajesh Palit
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Subject: Submission of the Capstone Project Report on "Multi-Tasking Autonomous Device"

Dear Sir,

With due respect, we are pleased to submit our Capstone Project Report on **"Multi-Tasking Autonomous Device"** as part of our BSc program. The project gave us valuable practical experience, and we were able to apply what we have learnt in the classroom to a real engineering problem. We have tried our best to meet all the requirements of the report.

We would be highly obliged if you would kindly receive this report and give us your valuable judgment. We hope you find it useful and informative.

Sincerely yours,

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Declaration

This is to declare that this project is our original work. No part of it has been submitted elsewhere, partially or fully, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous work presented in this report has been properly acknowledged and cited, and the plagiarism policy stated by the supervisor has been maintained.

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Abstract

Robotics is among the fastest-growing fields of engineering, and the COVID-19 pandemic has made sanitation and sterilisation a part of everyday life. The pandemic also increased demand for automated cleaning and disinfecting devices that spare people from exposure to infectious surfaces. This report presents the design, implementation and testing of a low-cost **Multi-Tasking Autonomous Device** for enclosed indoor spaces, built around an Arduino microcontroller on a four-wheel chassis.

The device combines several functions: obstacle detection and avoidance with a servo-mounted ultrasonic sensor; ultraviolet (UV) LED disinfection that is switched off automatically when a PIR sensor detects human motion; floor cleaning with a vacuum unit; a smoke-based fire alarm; light and temperature readings shown on an LCD; fingerprint-based user authentication; and remote surveillance through a Wi-Fi mini camera. The circuits were verified in Tinkercad simulation and then built as a hardware prototype. In our tests the obstacle-avoidance module prevented collisions in about 97% of trials. The total cost of the prototype was approximately BDT 8,000, and a commercial reproduction is estimated to cost no more than BDT 10,000.

Keywords: autonomous robot, obstacle avoidance, ultrasonic sensor, PIR sensor, UV disinfection, fire alarm, fingerprint authentication, Arduino.

Table of Contents

Chapter 1: Introduction **1**
1.1 Background and Motivation 1
1.2 Purpose and Goal of the Project 1
1.3 Organization of the Report 3
1.4 Problem Statement 3
Chapter 2: Research Literature Review **5**
2.1 Existing Research and Limitations 5
2.2 Proposed Solution 6
2.3 Solution Assessment & Project Goals 7
2.4 Design Alternatives 9
2.5 Technical Design: Module Level 9
2.6 Technical Design: System Level 10
2.7 Required Skills 10
Chapter 3: Methodology **11**
3.1 Technology Platform 11
3.2 Hardware and Software Components 11
3.3 Work Breakdown Structure 17
3.4 Related Work 18
3.5 System Design 19
3.6 Simulation and Hardware Implementation 23
Chapter 4: Results, Analysis and Discussion **28**
4.1 Result and Discussion 28
4.2 Feasibility Study 30
4.3 Troubleshooting 30
4.4 Limitations 30
4.5 Future Development 31
4.6 Project Demonstration Review 32
4.7 Lessons Learnt 32
Chapter 5: Impacts of the Project **34**
5.1 Societal, Health, Safety, Legal and Cultural Impact 34
5.2 Environmental and Sustainability Impact 34
Chapter 6: Project Planning and Budget **35**
6.1 Financial Plan and Costs 35
6.2 Project Timeline 36
Chapter 7: Complex Engineering Problems and Activities **37**
7.1 Complex Engineering Problems (CEP) 37
7.2 Complex Engineering Activities (CEA) 38
Chapter 8: Conclusions **39**
8.1 Summary 39
8.2 Limitations 39
8.3 Future Improvement 40
8.4 Project Demonstration Review 40
8.5 Project Achievements 40
8.6 Lessons Learnt 41
Appendix A: References **42**
Appendix B: Codes **44**

List of Figures

Figure 2.1: Obstacle-avoidance concept: the ultrasonic sensor detects an obstacle and the vehicle turns towards the clearer side. 7
Figure 2.2: UV sterilisation, fire alarm system and vacuum cleaning modules. 7
Figure 2.3: Fingerprint scanning (left) and surveillance and monitoring (right). 8
Figure 3.1: Arduino Uno board. 11
Figure 3.2: L293D motor driver. 12
Figure 3.3: Jumper wires. 12
Figure 3.4: Double-sided tape. 13
Figure 3.5: Battery holder. 13
Figure 3.6: HC-SR04 ultrasonic sensor. 13
Figure 3.7: SG90 mini servo motor. 14
Figure 3.8: Photoresistor (LDR). 14
Figure 3.9: LM35 temperature sensor. 14
Figure 3.10: Four-wheel robot chassis. 15
Figure 3.11: Working principle of the ultrasonic sensor: a 10 µs trigger pulse starts an 8-cycle burst, and the echo pulse width is proportional to the distance. 19
Figure 3.12: Block diagram of the obstacle-avoidance module: battery, ultrasonic sensor (HC-SR04), Arduino Uno, motor controller (L293D) and motors. 20
Figure 3.13: Wiring of the motor-driver shield with the DC motors, servo motor, ultrasonic sensor, switch and 7.4 V battery. 20
Figure 3.14: Arduino Uno connected to the PIR sensor and LED. 20
Figure 3.15: Reference gas-leak detection flowchart (adapted from [13]); SMS notification can be added with a GSM module. 21
Figure 3.16: Arduino connected to a buzzer and indicator for the alarm circuit. 21
Figure 3.17: Block diagram of the complete system, including the vacuum cleaner and UV light. 22
Figure 3.18: Circuit for biometric authentication (fingerprint scanner connected to the Arduino). 22
Figure 3.19: Light and temperature reading on the LCD screen mounted on the prototype. 23
Figure 3.20: Tinkercad schematic of the PIR sensor and LED circuit. 24
Figure 3.21: System block diagram (without the vacuum module). 24
Figure 3.22: Circuit diagram of the Arduino, L293D motor driver, ultrasonic sensor and motors. 24
Figure 3.23: Flow chart of the obstacle-avoidance decision logic. 25
Figure 3.24: Tinkercad simulation of the PIR sensor, smoke sensor, buzzer and LCD with an Arduino Uno. 26
Figure 3.25: Tinkercad simulation of the obstacle-avoiding vehicle with ultrasonic sensors and motors. 26
Figure 4.1: Top view of the final product. 28
Figure 4.2: Front-on view of the final product. 28
Figure 4.3: Side-on view of the final product. 29
Figure 8.1: Front-on close-up of the multi-tasking autonomous device. 39

List of Tables

Table I: Summary of existing research and its limitations 5
Table II: Design alternatives considered 9
Table III: Purpose of the main components 16
Table IV: Additional hardware 16
Table V: Comparison of this project with related devices 18
Table VI: Modules of the final prototype 29
Table VII: Problems faced and solutions 30
Table VIII: Financial plan / budget and costs (base hardware) 35
Table IX: Weekly progress for EEE499A 36
Table X: Weekly progress for EEE499B 36
Table XI: Complex engineering problems (CEP) 37
Table XII: Complex engineering activities (CEA) 38

Chapter 1: Introduction

1.1 Background and Motivation

The rapid spread of the highly infectious COVID-19 virus caused far-reaching social and economic changes and forced people to look beyond social distancing and government guidelines for ways to reduce the threat of infectious pathogens. Sanitation and sterilisation have since become an inseparable part of daily life, and the growth of a germ-conscious mindset has increased interest in automated cleaning, sanitising and disinfecting devices. However, for communicable diseases spread by airborne microbes, it is unwise for people to take part physically in the sterilisation process, because there is a considerable risk of infection from the contaminated space.

Necessity is the mother of invention, and this situation calls for a versatile, efficient and autonomous device that can sterilise a space with relative ease and without direct human involvement. Robots are already used in several applications where people would be exposed to risk, so we set out to build a microbe-inactivating UV disinfection module that can be used in hospitals as well as in apartments and offices.

Fire and air quality are further concerns. A study of burn epidemiology in Bangladesh reports that nearly 3,000 people die each year from burn-related injuries, and fire alarm systems play an important role in warning occupants and thereby saving lives. Poor indoor air is another major health burden: indoor air pollution is estimated to cause about 3.8 million deaths each year, because particulate matter and harmful gases can lead to asthma, reduced lung function and even cancer. Leaks of combustible or toxic gases can also cause explosions, poisoning and oxygen displacement, so early detection lets occupants evacuate and limits damage.

A vacuum-cleaning module makes the device attractive to people who want help with household cleaning: it sweeps the floor, collects dirt on the way and can be stopped at will. Finally, access control and remote monitoring add security, so that only authorised people operate the device and the owner can check the space from a distance.

The novelty of this project lies in combining these functions in one compact, low-cost robot-like device. It avoids obstacles, switches off its UV LEDs when human motion is detected nearby, disinfects an enclosed space without human intervention, vacuum-cleans the floor, raises an alarm on smoke, displays light and temperature, authenticates users by fingerprint and streams live video to a smartphone. Its uniqueness is this multi-functionality at a price that ordinary households and small institutions can afford.

1.2 Purpose and Goal of the Project

The purpose of the project is to design, build and test an autonomous multi-tasking device that moves around an indoor space on its own while performing useful tasks, and that warns occupants of hazards. By using sensors, motors and a microcontroller, the device can clean the

floor and disinfect surfaces without human intervention, which saves time and effort and adds a layer of safety. As a home or office appliance it offers a more convenient way to handle routine cleaning, and its safety features give users peace of mind. We chose robotics because it is a rapidly expanding field and a direct way to modernise everyday technology, especially in developing countries where affordable automation is scarce.

The specific goals are to:

- build an Arduino-based mobile platform that navigates an unknown environment and avoids obstacles;
- disinfect enclosed spaces with UV LEDs while switching them off automatically when people are nearby;
- clean the floor with an on-board vacuum module;
- detect smoke and raise an audible alarm;
- measure and display light and temperature;
- restrict use to authorised people through fingerprint authentication;
- allow remote viewing of the surroundings through a Wi-Fi mini camera; and
- keep the cost low enough for homes, clinics and offices.

The functions are described briefly below.

Obstacle avoidance

The obstacle-avoidance system lets the device travel through an unknown environment without collisions: it detects an obstruction in its route, avoids it and continues. Common approaches to robot navigation include wall following, edge detection and line following. Edge-detection-based avoidance is widely used but requires the robot to stop in front of an obstacle to assess it accurately. Collision avoidance in mobile robots ranges from simple algorithms that merely halt the robot to more advanced ones that detour around the obstacle, which need obstacle recognition and some quantitative assessment of its size. Our device takes a middle path: it stops, scans both sides with a servo-mounted ultrasonic sensor and turns towards the side with more free space.

UV disinfection

The UV module is intended to sanitise indoor spaces without chemical disinfectants or direct human involvement. UV-C radiation (200–280 nm) is well known for its germicidal effect on bacteria and viruses and is used in hospital disinfection robots [1], [2], [5]. Because UV radiation is harmful to skin and eyes, the UV LEDs are coupled with a PIR motion sensor so that they switch off whenever a person moves nearby.

Vacuum cleaning

The vacuum module uses a small DC motor and fan to draw dust and debris through a suction opening into a collection chamber. The motor is powered from the robot's supply, while the ultrasonic sensor steers the device around obstacles so that it covers the floor automatically [11], [12].

Fire alarm

A smoke sensor is read by the Arduino. When the reading crosses a threshold, the Arduino sounds a buzzer to alert the occupants. The same principle can be extended with GSM-based SMS notification [13] and, in a later version, with automatic shut-off or fire-suppression mechanisms.

Biometric authentication, environmental readings and surveillance

A fingerprint scanner connected to the Arduino provides an authentication layer so that only enrolled users can interact with the device. An LDR and a temperature sensor measure ambient light and temperature, which are shown on an LCD. A Wi-Fi mini camera lets users watch the surroundings from a smartphone or other connected device.

1.3 Organization of the Report

The rest of the report is organised as follows:

- **Chapter 2** reviews the research literature, presents the proposed solution and goals, and describes the module-level and system-level design.
- **Chapter 3** describes the methodology: the technology platform, components, work breakdown, related work, system design, simulation and hardware implementation.
- **Chapter 4** presents the results, feasibility study, troubleshooting, limitations, future development, demonstration review and lessons learnt.
- **Chapter 5** discusses the impact of the project on society, health, safety, law, culture, the environment and sustainability.
- **Chapter 6** gives the financial plan and the project timeline.
- **Chapter 7** relates the project to complex engineering problems and activities.
- **Chapter 8** concludes the report with a summary, limitations, future improvements and achievements.
- **Appendices** list the references (Appendix A) and the Arduino code (Appendix B).

1.4 Problem Statement

Cleaning and maintaining indoor spaces is time-consuming and labour-intensive, and it becomes dangerous when the space must also be disinfected. Hazards such as fire or gas leaks can go unnoticed and call for quick, decisive action. Existing technologies address these needs

only separately: robotic vacuum cleaners do not disinfect or monitor hazards, UV disinfection robots are expensive and rarely clean floors, and smoke detectors are fixed devices that cannot move through a space.

The multi-tasking autonomous device aims to close this gap by providing one affordable platform that navigates on its own, cleans and disinfects, restricts access to authorised users, monitors the environment and warns occupants of danger. Automating these tasks frees up time and resources, and the safety features help occupants react quickly in an emergency.

Several challenges must be addressed to achieve this: integrating multiple sensors and control tasks on a limited microcontroller, supplying enough power for the motors, UV LEDs and vacuum motor from a small battery, and ensuring that the device operates safely and reliably in different indoor environments. Meeting them requires careful design, simulation, testing and iteration.

Chapter 2: Research Literature Review

2.1 Existing Research and Limitations

Studying related work shows which areas need more attention and where progress is possible, so we reviewed a number of research papers and projects before designing our device. The main findings are summarised in Table I and discussed below.

Table I: Summary of existing research and its limitations

Ref.	Work	Limitation / relevance to this project
[1]	Robotic UVGI platform for disinfecting radiology rooms; can work alongside staff in cramped rooms and reveals high microbial load in hard-to-reach places.	Designed for hospital radiology; not a general low-cost solution.
[2]	ICRA 2021 paper on path planning for regular UV-C irradiation of indoor environments; efficacy shown in simulation and a prototype run.	Does not appear portable or cost-effective.
[3]	Review of service robots for disinfection in medical institutions.	Robots are expensive and used mainly by professionals in hospitals and clinics.
[4], [10]	Basic concepts and speed control of obstacle-avoiding Arduino robot cars.	Navigation only; no disinfection or safety functions.
[5]	Disinfection robot with a mobile platform, six-axis arm and UV-C LED array, with remote control and path planning.	Not portable and very expensive.
[6]	Interview with the CEO of Fetch Robotics on adding disinfection payloads to mobile robot platforms during COVID-19.	Shows demand for flexible platforms; UV disinfection is not discussed in detail.
[7]	Obstacle-avoiding robot using IR and PIR sensors.	Navigation only.
[8], [9]	Evaluation of UV disinfection robots for hospitals; UV sterilisation robot.	[8] notes that wavelength and exposure time must be matched to the target pathogen and that one size does not fit all.
[11], [12]	Arduino-based floor cleaner and vacuum cleaner robot.	Cleaning only.
[13]	SMS-based fire alarm and gas-leak detector using Arduino and GSM.	Fixed installation; not mobile.
[14], [15]	IoT-based surveillance and monitoring systems.	Monitoring only.

Existing UV disinfection robots are effective but generally costly, bulky or limited to hospitals, so they are out of reach for homes and small offices. The ESCMID-related discussion in [8] points out that more work is needed on the wavelength and exposure time required to inactivate

a given pathogen, and that robots must communicate with their environment. Our project does not try to settle this question; it aims at an inexpensive, compact and portable device that combines UV disinfection with cleaning and safety features.

Robotic vacuum cleaners (often called “Robovacs”, with the Roomba as the best-known example) combine a limited vacuum system with sensors, robotic drives and programmable cleaning routines. Contemporary fire alarm systems detect events that may lead to a fire using smoke, heat or carbon-monoxide sensors and automatically pass the signal on to an alarm panel. Monitoring and surveillance systems [14] are more intricate but not far off from what a small Wi-Fi camera can offer.

For system control we looked at IoT-based control [15], GSM [13] and remote control with an antenna [16]. GSM (Global System for Mobile communication) is a digital mobile network used widely around the world, and the Internet of Things (IoT) refers to the network of connected devices and the technology that lets them communicate with each other and with the cloud. In practice, the choice for us depended on our technical skills and the resources available, so we kept the control system simple and Arduino-based.

2.2 Proposed Solution

In our system, a single ultrasonic sensor sweeps left and right on a servo motor, and the controller uses a predefined algorithm to decide where the vehicle should move next. We tried to make the system as cost-effective as possible. The initial version has a front-facing sensor to demonstrate the concept; rear and side sensors can be added according to a client’s needs. We preferred the ultrasonic sensor because it gives reliable distance data in unfamiliar environments and does not depend on lighting. If faster or more accurate processing is required and cost permits, the Arduino can be replaced by a Raspberry Pi or a custom microcontroller; initially we use the Arduino.

The proposed device is a compact and portable autonomous platform that avoids obstacles, switches off its UV LEDs when human motion is nearby, disinfects an enclosed space without human intervention, vacuum-cleans the floor and raises an alarm when it detects smoke. It also shows light and temperature, authenticates users by fingerprint and offers remote viewing. It consists of the following main elements:

- **Microcontroller:** an Arduino board receives and processes input from the sensors and controls the motors and other components.
- **Sensors:** an ultrasonic sensor for obstacles, a PIR sensor for human motion, a smoke sensor, an LDR and a temperature sensor.
- **Actuators:** DC motors drive the wheels and the vacuum fan, and a servo motor rotates the ultrasonic sensor.
- **User interface:** a fingerprint scanner for authentication and an LCD for readings and messages, plus a Wi-Fi mini camera for remote viewing.

- **Power supply:** a rechargeable battery with regulation to 5 V for the logic circuits.
- **Control system:** the Arduino programs that let the device navigate, avoid obstacles, control the UV LEDs and raise alarms.

By combining these elements in a single device, the system offers a more efficient way to clean indoor spaces while also improving safety and security.

2.3 Solution Assessment & Project Goals

The key requirement of the project is the safety of the device itself: it must detect an obstacle, stop accelerating and brake, and then decide where to move next. We assessed the solution design in terms of alignment and quality. The system was installed on a demonstration toy car without a dedicated braking system. It synced well with the car and worked properly: when an obstacle appears within the predefined range, the car stops accelerating and then turns left or right depending on which side has the larger distance. This protects the device from collisions, and the project meets the requirements of the stakeholders and the proposed solution.



Figure 2.1: Obstacle-avoidance concept: the ultrasonic sensor detects an obstacle and the vehicle turns towards the clearer side.

The functional modules are described below.



Figure 2.2: UV sterilisation, fire alarm system and vacuum cleaning modules.

A) **Sterilisation:** the system is designed to sterilise an enclosed place. Equipped with UV lights, it is intended to inactivate infectious microbes and pathogens such as viruses, bacteria, fungi and spores on exposed surfaces. Effectiveness depends on the UV wavelength, intensity, distance and exposure time.

B) **No human intervention:** the device moves across the floor of a closed space without human intervention, reducing the risk of contracting communicable pathogens. Autonomous technology enriches automated systems with sensors, artificial

intelligence and analytics so that they can make independent decisions from the data they collect.

C) **Obstacle avoidance:** the device detects obstacles automatically and redirects its path to avoid collisions. In robotics, obstacle avoidance means satisfying a control objective subject to non-collision position constraints. IR sensors respond faster than ultrasonic sensors, but ultrasonic sensors are the better choice for small, relatively slow devices like ours.

D) **Switch-off near human motion:** the UV lights are switched off automatically when a person moves nearby, to avoid exposing skin and eyes to UV radiation. A PIR sensor detects human presence from the change in infrared radiation across its detection zones, so it cannot reliably detect a completely stationary person (a false negative). Nearby human motion is, however, detected clearly, which is enough to switch the UV LEDs off.

E) **Vacuum cleaning:** a motor-driven suction system collects dust and debris through a suction opening and carries it into a storage container inside the device.

F) **Fire alarm:** when the smoke sensor reading crosses its threshold, the controller activates a buzzer to alert the occupants.

G) **Biometrics:** a fingerprint scanner is attached to the microcontroller. Users enroll their fingerprints, which are stored as templates; during authentication the scanner compares a scanned finger with the stored templates and reports a match only for enrolled users.

H) **LCD screen:** a Liquid Crystal Display gives visual feedback such as environmental readings (light and temperature), task status and authentication prompts.

I) **Surveillance and remote monitoring:** a Wi-Fi mini camera streams live video to smartphones and other devices, so homeowners and office managers can check their property remotely.

J) **Power supply:** a reliable power source is needed for uninterrupted operation; the device uses rechargeable batteries.



Figure 2.3: Fingerprint scanning (left) and surveillance and monitoring (right).

2.4 Design Alternatives

Three design alternatives were considered (Table II). The first is to use a Raspberry Pi instead of the Arduino. Both can run the same logic, but the Raspberry Pi has more processing power and responds more smoothly; it also permits a larger power supply and balanced weight distribution, at higher cost and power consumption. The second is to remove the servo motor and moving head and use two additional sonar sensors on the left and right instead. The side sensors would be read only when the front sensor detects a nearby obstacle, and the longer obstacle-free side would be chosen. The third is to use IR sensors in place of ultrasonic sensors.

Table II: Design alternatives considered

Alternative	Advantage	Drawback
Raspberry Pi instead of Arduino Uno	More processing power; smoother response	Higher cost and power consumption; more complex to set up
Two extra side sonar sensors instead of a servo-mounted sensor	No moving part; both sides measured at once	More sensors (cost and wiring); extra code logic
IR sensors instead of ultrasonic	Faster response time	Less suitable for small, slow devices; ultrasonic gives more reliable range data

2.5 Technical Design: Module Level

At the module level the system is highly modular: each component has its own function, and combining them makes the system work. The design is highly cohesive. Modules such as the sonar sensor, the wheels and the servo motor show **functional cohesion**, because every element needed for a single computation is contained in the component.

The whole system also shows **sequential cohesion**, in which the output of one element becomes the input of the next. The motors run when the microcontroller sends the signal and stop when the sonar sensor detects an obstacle; the servo motor runs only when there is an obstacle, to measure the right and left distances. The wheels operate on the same input data, which is an example of **communicational cohesion**. There is no sign of coincidental cohesion.

The modules are loosely coupled. They are mostly connected by **data coupling**, in which components are independent and communicate only through data, and by **external coupling**, in which modules depend on hardware or software outside the program being developed. There is no **content coupling**, so one module cannot modify the data of another.

2.6 Technical Design: System Level

At the system level, the physical design describes how data is entered, verified, processed and acted upon as output. The sonar sensor continuously supplies data (the measured distance, a number) to the microcontroller, which processes it and outputs commands, for example a Boolean decision on whether the motors and servo should run. The PIR, smoke, light and temperature sensors and the fingerprint scanner provide further inputs, and their outputs control the UV LEDs, buzzer and LCD. The system is sequential: each stage depends on the result of the previous one.

2.7 Required Skills

The skills required to build the system are modest. A clear understanding of the requirements and the expected output is essential. The control algorithms were written in the Arduino language, which is based on C++; Python or C could have been used as well. We learnt how the microcontroller is wired and how its functions work in order to establish the connections, and our basic knowledge of electronics made this easy to pick up. Familiarity with circuit simulation in Tinkercad was also helpful for testing before assembly.

Chapter 3: Methodology

3.1 Technology Platform

The technology platform of the project consists of hardware and software components.

Hardware: the main components are the microcontroller board; the sensors (ultrasonic, PIR, smoke, LDR and temperature); DC motors with a motor driver; a servo motor; the vacuum-cleaner mechanism; UV LEDs; a fingerprint scanner; an LCD; a Wi-Fi mini camera; a rechargeable battery; and the mechanical parts required for movement.

Software: the microcontroller is programmed in the Arduino Integrated Development Environment (IDE). The programs interpret the sensor signals and control the other components. Tinkercad was used for circuit simulation, Multisim for circuit analysis and MATLAB for computation.

Overall, the platform integrates these hardware and software parts into an automated system able to perform several tasks without human intervention.

3.2 Hardware and Software Components

Arduino Uno

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P and developed by Arduino.cc. It has 14 digital input/output pins, of which 6 can be used as PWM outputs, and 6 analog inputs. It also has a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It is widely used in electronic projects because it is low-cost, flexible and easy to program.



Figure 3.1: Arduino Uno board.

Motor driver

A motor driver is an interface between the motors and the control circuit. The microcontroller cannot supply the current that motors need, so the driver takes low-power control signals and switches the motor supply accordingly, controlling speed and direction. L293D and ULN2003 are two of the most common motor-driver ICs. In this project the L293D motor-driver shield is used with the AFMotor library to drive four DC motors and a servo.



Figure 3.2: L293D motor driver.

Connecting wires (jumpers)

Jumper wires have connector pins at each end, so they can connect two points without soldering. They are typically used with breadboards and other prototyping tools so that a circuit can be changed easily; the colours of the wires carry no meaning. They come in male-to-male, male-to-female and female-to-female versions, which differ in the end points: male ends have a protruding pin and plug into things, while female ends accept a pin. Male-to-male wires are the most common.



Figure 3.3: Jumper wires.

Double-sided tape

Double-sided tape is a pressure-sensitive tape coated with adhesive on both sides, used to stick two surfaces together. In this project it was used to fix lightweight modules and sensors to the chassis.



Figure 3.4: Double-sided tape.

Battery holder

A battery holder is a compartment that holds one or more batteries and makes electrical contact with their terminals. The holder used here connects four AA cells in series to power the project.



Figure 3.5: Battery holder.

Sonar (ultrasonic) sensor

An ultrasonic (sonar) sensor measures distance by sending a short burst of ultrasound and timing the echo that returns from an object. We use the HC-SR04 module, which measures distances from 2 cm to 400 cm with an accuracy of up to 3 mm and a measuring angle of about 15°. It has four pins: VCC, Trig, Echo and GND.



Figure 3.6: HC-SR04 ultrasonic sensor.

Servo motor

A servo motor is a rotary or linear actuator that allows precise control of angular or linear position, velocity and acceleration. It consists of a motor coupled to a sensor for position feedback. Servo motors come in two basic types, AC and DC. In this project a small SG90 servo rotates the ultrasonic sensor to the left, right and centre.



Figure 3.7: SG90 mini servo motor.

Photoresistor (LDR)

A photoresistor, also called a photocell, light-dependent resistor (LDR) or photoconductive cell, is a passive component whose resistance decreases as the light falling on its sensitive surface increases. It is used here to measure ambient light.



Figure 3.8: Photoresistor (LDR).

Temperature sensor (LM35)

The temperature sensor is a three-pin analog device whose output voltage is proportional to temperature. The Arduino reads the voltage on an analog input and converts it to degrees Celsius.



Figure 3.9: LM35 temperature sensor.

Chassis

The chassis is the main support structure (frame) of the vehicle. We used a four-wheel-drive robot chassis kit with DC motors and wheels, which carries all the electronics and bears the mechanical stresses.



Figure 3.10: Four-wheel robot chassis.

Other components

- **PIR sensor (HC-SR501):** detects the change in infrared radiation caused by a moving person and outputs a digital HIGH signal.
- **UV LEDs:** provide the ultraviolet light used for disinfection.
- **Smoke sensor:** an analog gas/smoke sensor read on analog pin A5; a buzzer sounds when the reading exceeds a threshold.
- **Fingerprint scanner:** an optical scanner that captures a fingerprint image, converts it to a template and compares it with stored templates; it talks to the Arduino over a serial link and is driven with the Adafruit Fingerprint library.
- **16×2 LCD:** displays the temperature and light readings; driven with the LiquidCrystal library.
- **Wi-Fi mini camera:** a compact camera module that streams video over Wi-Fi to a smartphone or other device.

Software tools

- **Arduino IDE** contains a text editor, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.
- **MATLAB** is a high-performance language for technical computing that integrates computation, visualisation and programming in an easy-to-use environment.
- **Multisim** integrates industry-standard SPICE simulation with an interactive schematic environment to visualise and analyse circuit behaviour.
- **Tinkercad** is a browser-based tool for simulating Arduino circuits and code.

Table III summarises the purpose of the main components, and Table IV lists the additional hardware used.

Table III: Purpose of the main components

Component	Purpose in this project
Arduino Uno	Main controller. It connects to a computer by USB for powering and programming and runs the programs written in the Arduino language.
Motor driver (L293D)	Interface between the motors and the control circuit. The Arduino outputs only 5 V at a few milliamps, so a driver is needed to switch the higher motor supply.
Connecting wires (jumpers)	Connect two points of the circuit without soldering.
Sonar sensor (HC-SR04)	Measures distance and detects obstacles. The Trig pin sends the ultrasonic pulse and the Echo pin receives the reflection.
Servo motor (SG90)	Rotates the ultrasonic sensor left and right so the distance on each side can be compared.
5 V DC motors	Drive the wheels and move the vehicle.
PIR sensor and UV LEDs	The PIR sensor detects nearby human motion so that the UV LEDs can be switched off.
Smoke sensor and buzzer	Detect smoke and sound the alarm.
LDR, temperature sensor and LCD	Measure ambient light and temperature and display them.
Fingerprint scanner	Authenticates enrolled users.
Wi-Fi mini camera	Provides a live video stream for remote monitoring.
Battery	Supplies power to the motors and electronics; a rechargeable pack allows repeated use.
Wheels	Reduce friction and allow the vehicle to move and turn.
Chassis	The main support structure (frame) that carries the components.

Software used: (a) MATLAB, (b) Multisim, (c) Arduino IDE for Windows, (d) Tinkercad.

Table IV: Additional hardware

No.	Component	Image
1	HC-SR04 ultrasonic module	
2	L293D motor driver	

No.	Component	Image
3	5 V N20 motors and mounting brackets	
4	N20 motor wheels	
5	UV LEDs	
6	7.4 V Li-ion battery	
7	7805 voltage regulator	
8	330 Ω resistor	
9	Breadboard	
10	Polyester (perf) board	
11	Castor wheel	
12	Arduino Pro Mini (ATmega328P, 5 V / 16 MHz)	
13	PIR sensor HC-SR501	

3.3 Work Breakdown Structure

The work was divided into the following modules:

A) **Autonomous obstacle-avoiding vehicle:** detects obstructions in its path and avoids them. The Arduino is the central control unit that processes data and sensor signals and sets the parameters that guide the vehicle. The module is fully automatic and needs no human control when placed in an unfamiliar environment.

B) **UV-LED disinfection with human-presence cut-off:** the UV LEDs sterilise an enclosed place, and a PIR sensor switches them off when human motion is detected nearby so that skin and eyes are not exposed. Because PIR sensors respond to motion, a perfectly still person may not be detected; nearby motion, however, is detected clearly and is sufficient to switch the LEDs off.

C) **Fire alarm:** the smoke sensor sends an analog signal to the Arduino. When the reading exceeds its threshold, the Arduino sounds a buzzer; it could also trigger a siren or a fire-suppression action.

D) **Vacuum cleaning:** a suction motor and fan collect dust and debris and transport them to a storage container inside the device.

E) **Light and temperature reading:** an LDR and a temperature sensor are sampled by the Arduino and the values are shown on an LCD.

F) **Biometric authentication:** a fingerprint scanner is enrolled and queried through the Adafruit Fingerprint library.

G) **Surveillance:** a Wi-Fi mini camera is mounted on the device for remote viewing.

3.4 Related Work

The development of autonomous devices that perform several tasks is an active area of research and commercial development. The most relevant existing examples are:

- **iRobot Roomba:** a robotic vacuum cleaner that uses sensors to navigate around a room and avoid obstacles, and a dirt-detection sensor to find areas that need extra cleaning.
- **LG Hom-Bot:** an autonomous vacuum cleaner that uses a camera, ultrasonic sensors and infrared sensors to navigate a room and avoid obstacles.
- **UV disinfection robots:** commercial and research robots, such as those used in hospitals, carry UV-C lamps on mobile bases to disinfect rooms [1], [2], [5], [9]. They are effective but costly and do not clean floors.
- **Arduino-based prototypes:** academic projects that combine an Arduino with ultrasonic or IR sensors for obstacle avoidance [4], [7], [10], floor cleaning [11], [12] or fire and gas alarms [13].

Table V: Comparison of this project with related devices

Device	Obstacle avoidance	Floor cleaning	UV disinfection	Fire / smoke alarm
Robotic vacuum (e.g. Roomba, Hom-Bot)	Yes	Yes	No	No
UV disinfection robot	Yes	No	Yes	No
Fixed smoke detector	No	No	No	Yes
This project	Yes	Yes	Yes	Yes

This project builds on existing work and aims to bring these capabilities, together with fingerprint authentication, environmental readings and remote viewing, into one low-cost device.

3.5 System Design

The technical approach is to build an automated system around an Arduino microcontroller. Sensors respond to changes in the device's surroundings, and the controller activates the appropriate components. This section describes each module in turn.

3.5.1 Obstacle avoidance

The vehicle detects objects with an ultrasonic sensor. The system started with a single sensor, but a vehicle has blind spots on its left and right, so the sensor is mounted on a servo that sweeps both sides, limiting the chance of collision. The robot treats anything inside a particular range as an obstacle, avoids it and changes direction.

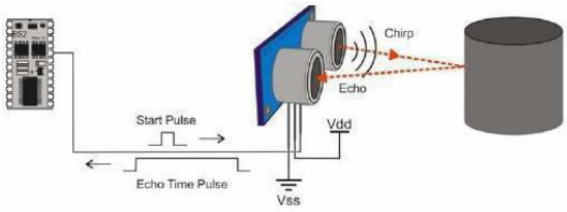


Figure 3.11: Working principle of the ultrasonic sensor: a 10 µs trigger pulse starts an 8-cycle burst, and the echo pulse width is proportional to the distance.

The sensor emits an ultrasonic pulse; the pulse reflects from a nearby object and returns as an echo. The Arduino measures the time between the pulse and the echo and computes the distance using the speed of sound (about 340 m/s). In the code, the distance in centimetres is the echo pulse width in microseconds divided by 58.82 (the round trip at 340 m/s). When the measured distance is 30 cm or less, the controller stops the motors, turns the servo to the right and then to the left to measure the distance on each side, and turns the vehicle towards the side with the larger distance. If no obstacle is detected, the vehicle moves forward. The decision logic is shown in the flow chart in Section 3.6.

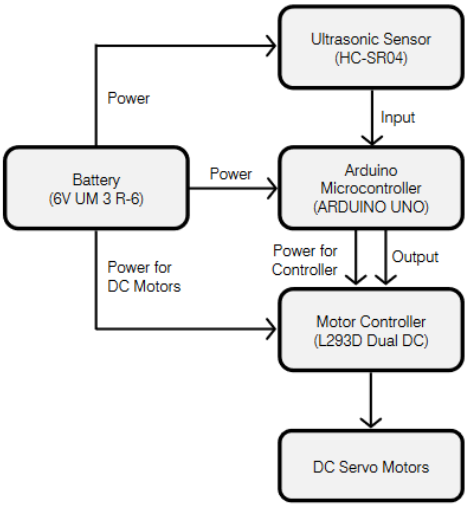


Figure 3.12: Block diagram of the obstacle-avoidance module: battery, ultrasonic sensor (HC-SR04), Arduino Uno, motor controller (L293D) and motors.

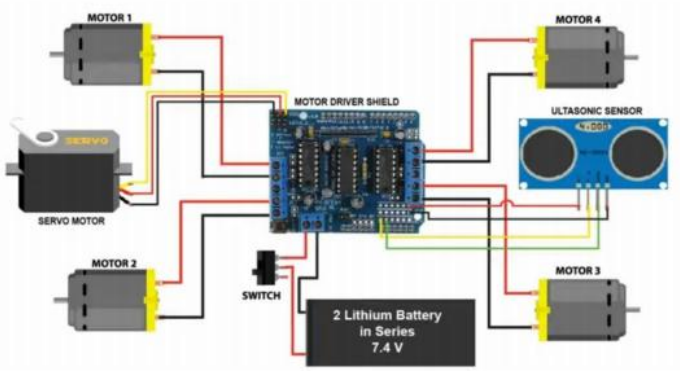


Figure 3.13: Wiring of the motor-driver shield with the DC motors, servo motor, ultrasonic sensor, switch and 7.4 V battery.

3.5.2 Human-motion detection and UV switching

A PIR sensor detects human motion in close proximity (about 25 cm in our tests) and sends a digital HIGH signal to the Arduino, which turns the UV LEDs off. The LEDs are turned on again once no motion is detected within that range.

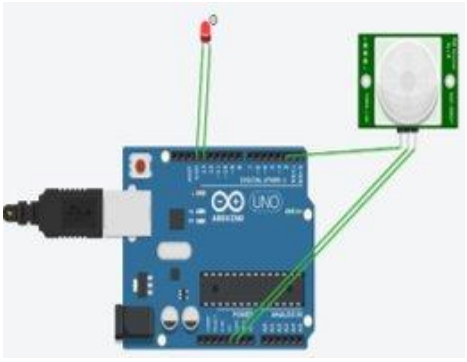


Figure 3.14: Arduino Uno connected to the PIR sensor and LED.

3.5.3 Fire alarm

Fire alarm systems are crucial for the safety of buildings and factories. They typically use a group of sensors that monitor for flames, smoke or gas leaks and trigger an alarm when one is detected. A common method is the infrared flame sensor, whose photodiode detects the infrared light emitted by a fire; an op-amp sets its sensitivity, and it gives a HIGH signal when a fire is detected, which the Arduino uses to sound a buzzer and light an LED [13]. In our prototype the Arduino reads an analog smoke sensor on pin A5; when the reading exceeds 200, the buzzer sounds a 1 kHz tone and a message is printed on the serial monitor. The threshold can be adjusted for the environment.

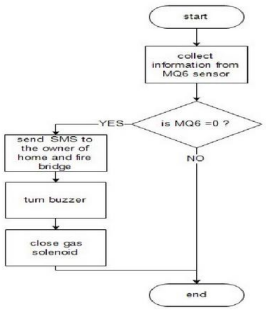


Figure 3.15: Reference gas-leak detection flowchart (adapted from [13]); SMS notification can be added with a GSM module.

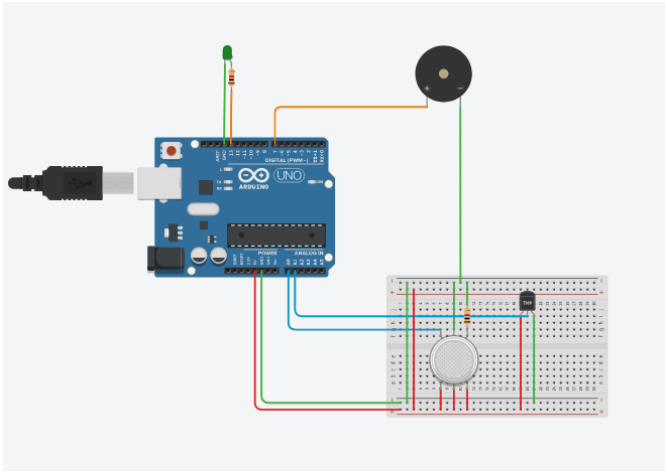


Figure 3.16: Arduino connected to a buzzer and indicator for the alarm circuit.

3.5.4 Vacuum cleaning

The vacuum module is based on a small portable vacuum cleaner. Such a cleaner has three main parts: a chamber that holds the dust, a middle section that contains the motor and fan, and a battery compartment. Its DC motor normally runs from 3 V (two 1.5 V AA cells) through a simple switch. In our design the original battery is removed, unnecessary parts are discarded and only the two-wire motor and fan are kept, powered from the robot's regulated supply.

To avoid falling down stairs, an IR proximity sensor can be mounted under the robot. It consists of an IR LED and a photodiode; the output changes when the emitted light is reflected from the floor, so a sudden change indicates an edge and the robot can stop and turn back. The ultrasonic sensors share a common ground and 5 V supply, with their trigger and echo pins connected to Arduino pins. The L293D enable pins and driver supply are connected to 5 V, since the motors are 5 V types. A 7.4 V Li-ion battery supplies the system, and an LM7805 regulator provides the 5 V rail for the Arduino, ultrasonic sensors, motor driver and motors. In the compact version the circuit is soldered on a small perfboard using headers for an Arduino Pro Mini, and the body is a round MDF robot (hard and water-repellent) with a circular opening for the vacuum unit, motors fixed with brackets, a castor wheel, ultrasonic sensors on the left, centre and right, and an external power switch.

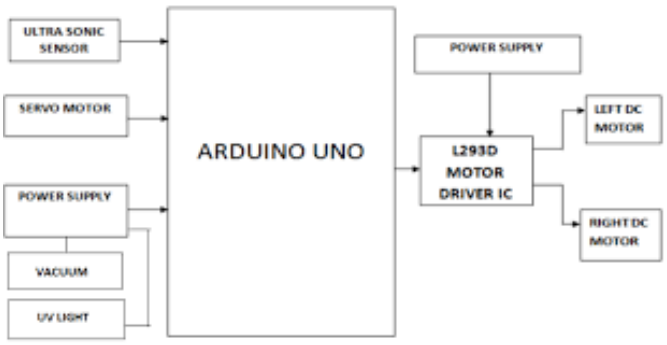


Figure 3.17: Block diagram of the complete system, including the vacuum cleaner and UV light.

3.5.5 Biometric authentication

Biometric authentication confirms a person's identity from distinctive physical traits. The device uses an Arduino, an optical fingerprint scanner and the Adafruit Fingerprint library. The scanner captures the fingerprint image and converts it to a digital template, and it exchanges commands and data with the Arduino over a serial link. The library handles enrolment, storage and verification: during enrolment a finger is scanned several times to create a template that is stored in the scanner's memory, and during authentication the scanned finger is compared with the stored templates. If a match is found, the program reports the template ID and a confidence value, which can be used to grant access to the device.

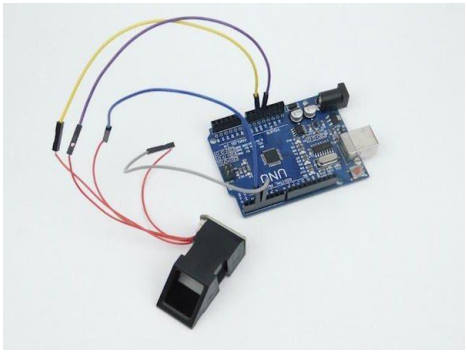


Figure 3.18: Circuit for biometric authentication (fingerprint scanner connected to the Arduino).

3.5.6 Light and temperature reading

The LDR is connected to an analog input, and the Arduino converts the voltage to a resistance value to decide whether the surroundings are dark or bright. The temperature sensor is read on another analog input, and the voltage is converted to degrees Celsius. Both values are shown on a 16×2 LCD driven with the LiquidCrystal library; the Arduino samples the sensors at fixed intervals, converts the readings and refreshes the display so that users can see the current conditions.

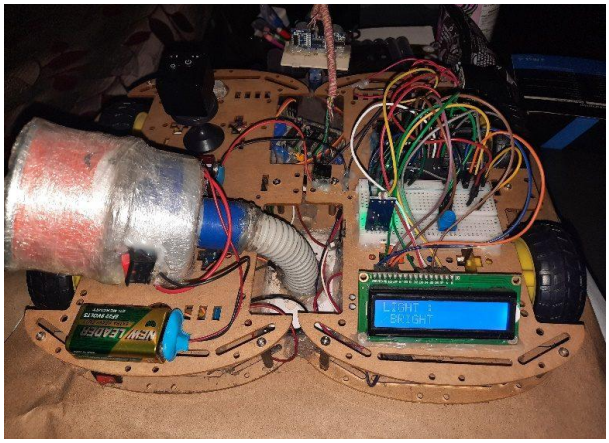


Figure 3.19: Light and temperature reading on the LCD screen mounted on the prototype.

3.5.7 Surveillance and monitoring

A Wi-Fi mini camera is mounted on the device. The camera module captures video and transmits it over a Wi-Fi network, so that users can view a live stream on a smartphone or other connected device through a mobile app or web interface and check the room from a distance. This adds security and convenience, since users can stay aware of their surroundings when they are not physically present.

3.6 Simulation and Hardware Implementation

Circuit description

The Arduino is connected to the motor driver, which controls the four motors that run the wheels (left front, left back, right front and right back motors on terminals M1–M4 of the driver). The ultrasonic sensor has four pins (VCC, Trig, Echo and GND), connected to the supply, A0, A1 and ground respectively; Trig works as an output and Echo as an input. The servo motor, which rotates the ultrasonic sensor, is connected to pin 10. Power comes from a Li-ion battery pack in series (7.4 V) connected to the power input of the board.

The PIR motion sensor has three pins (VCC, OUT and GND): VCC and GND are connected to the 5 V and ground pins of the Arduino and the output pin to pin 2. The positive end of the LED is connected to pin 13 and the negative end to ground.

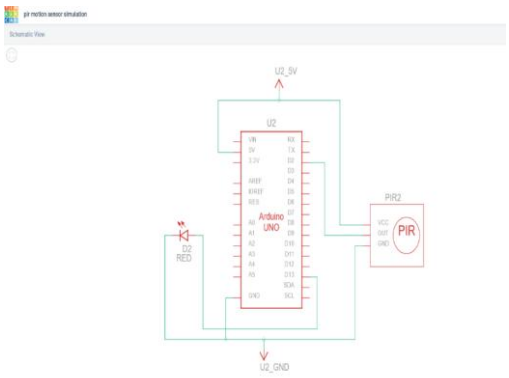


Figure 3.20: Tinkercad schematic of the PIR sensor and LED circuit.

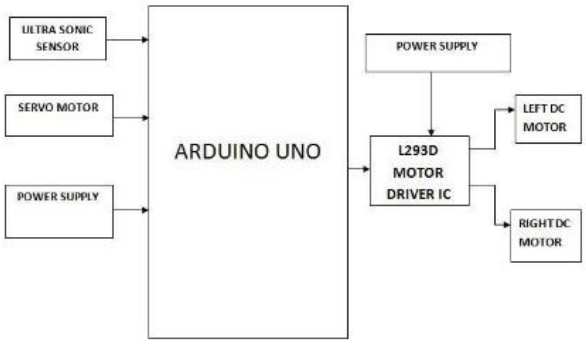


Figure 3.21: System block diagram (without the vacuum module).

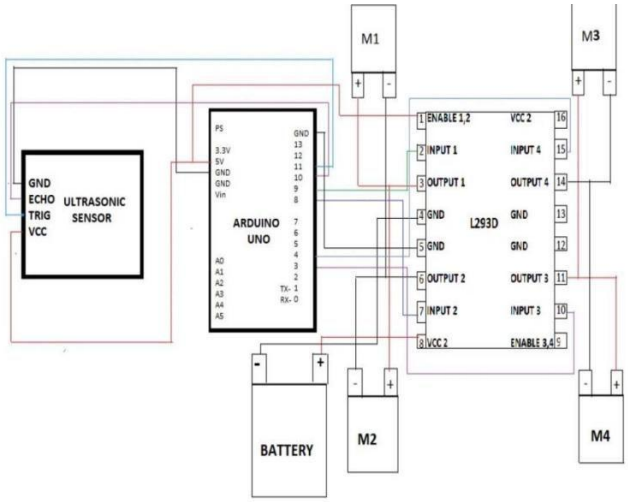


Figure 3.22: Circuit diagram of the Arduino, L293D motor driver, ultrasonic sensor and motors.

Operation

The ultrasonic sensor produces an echo signal when an obstacle is at close range (30 cm), and the Arduino reacts according to its program: it signals the L293D motor controller, which stops the motors immediately, as if a brake had been pressed. The servo then turns first to the right and then to the left, and the Arduino compares the two distances and signals the motor controller to turn towards the safer side. With the PIR sensor, a signal reaches the Arduino when human motion is detected, and the Arduino turns the UV LEDs off; they come back on automatically when no motion is detected. All of this is programmed in C++ using the Arduino IDE (Appendix B).

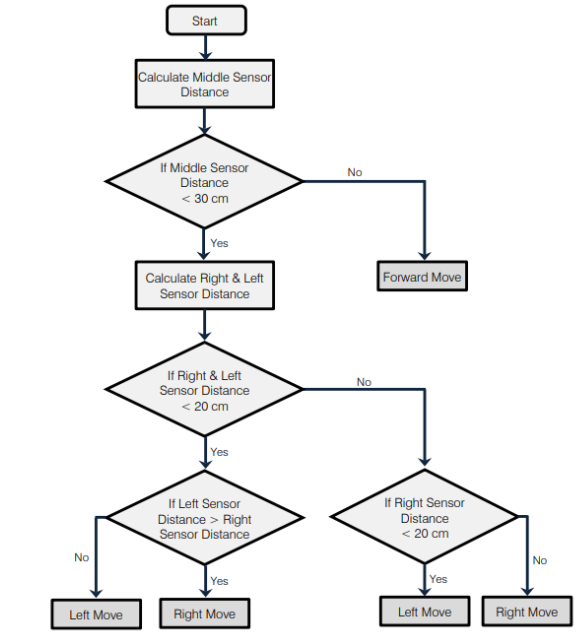


Figure 3.23: Flow chart of the obstacle-avoidance decision logic.

Simulation

Before assembling the hardware, the circuits were simulated in Tinkercad. The steps followed were:

1. **Select components:** choose parts similar to the ones to be used, such as an Arduino board, motors, LEDs, ultrasonic, PIR and smoke sensors and a power supply.
2. **Assemble the circuit:** drag the components into the workspace and connect them with virtual wires that match the intended physical layout.
3. **Configure properties:** set component properties such as the smoke-sensor sensitivity, the ultrasonic trigger distance and the motor voltage.
4. **Program the microcontroller:** upload the code for motor control, obstacle detection, motion detection and the alarm logic.
5. **Examine the findings:** run the simulation and check the motors, obstacle detection and alarm triggering; identify problems and areas that need improvement.
6. **Document:** keep screenshots and notes of key observations and conclusions.
7. **Iterate and refine:** adjust the circuit, component properties or program logic based on the results and repeat the simulation.

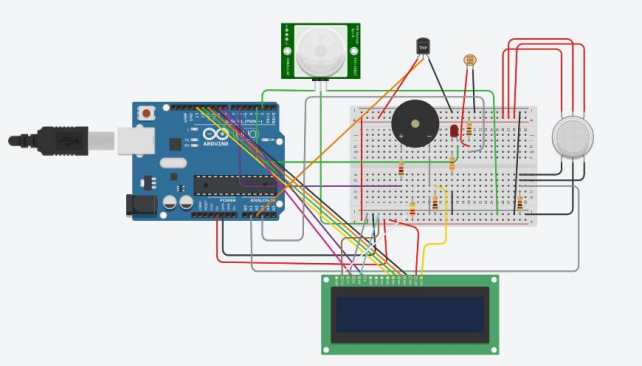


Figure 3.24: Tinkercad simulation of the PIR sensor, smoke sensor, buzzer and LCD with an Arduino Uno.

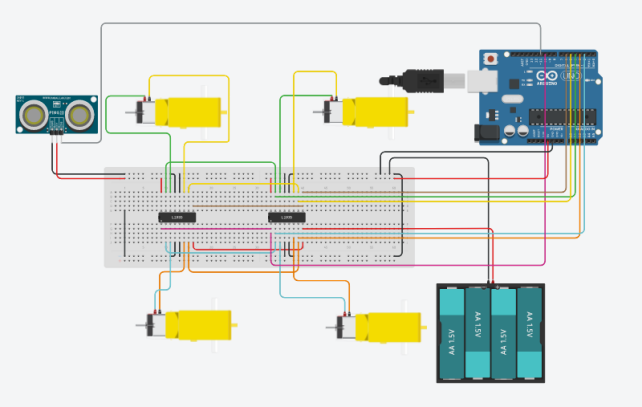


Figure 3.25: Tinkercad simulation of the obstacle-avoiding vehicle with ultrasonic sensors and motors.

Hardware implementation

The hardware prototype integrates the following parts on the chassis:

- **Microcontroller:** an Arduino acts as the brain of the system, processing sensor data and executing the algorithms for movement and task execution.
- **Sensors:** the ultrasonic sensor detects obstacles; the PIR sensor detects human motion; the smoke sensor detects smoke; the LDR and temperature sensor measure ambient conditions.
- **Actuators:** DC motors drive the wheels, a servo rotates the ultrasonic sensor and the vacuum motor creates suction. The motors are controlled by the microcontroller based on the sensor data.
- **UV disinfection:** UV LEDs built into the device emit the ultraviolet light used for disinfection.
- **Biometrics, LCD and camera:** the fingerprint scanner, the LCD and the Wi-Fi mini camera described in Section 3.5.
- **Power supply:** rechargeable batteries power the motors and electronics.

Each module was developed and tested as its own Arduino sketch (Appendix B). Because the sketches use overlapping pins (for example pins 2, 10 and 13), running all modules from one

controller requires reassigning pins or adding a second board; this is noted under future development. Accurate assembly, wiring and calibration were required for good performance.

Chapter 4: Results, Analysis and Discussion

4.1 Result and Discussion

In our tests the system detected obstacles and prevented collisions in about 97% of trials. Occasionally, when an obstacle appears suddenly in front of the device, the system fails to stop in time. There are two reasons: (i) the device is moving and accelerating constantly, and (ii) the controller needs a short time to process the sensor data after detecting the obstacle. Combined with the fact that our demonstration car has no dedicated braking system, so that stopping takes longer, these factors cause the car to hit a suddenly appearing obstacle. Apart from this issue, the system performs consistently.

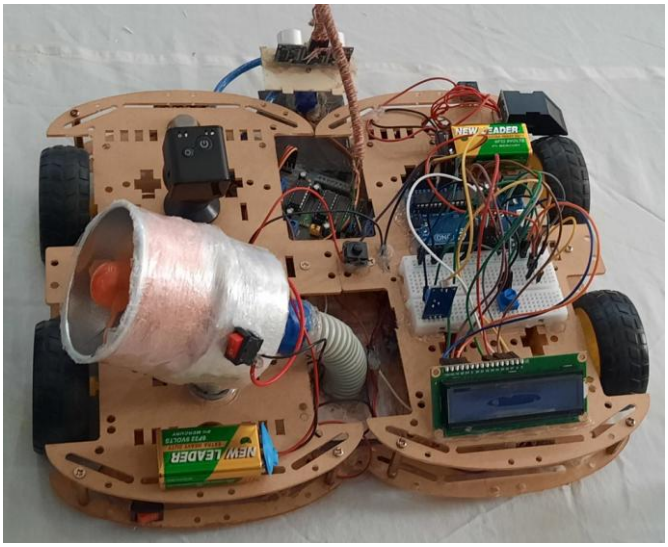


Figure 4.1: Top view of the final product.

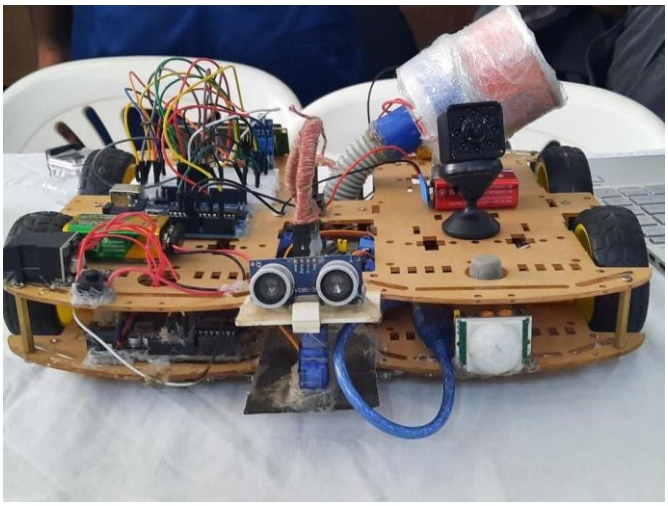


Figure 4.2: Front-on view of the final product.

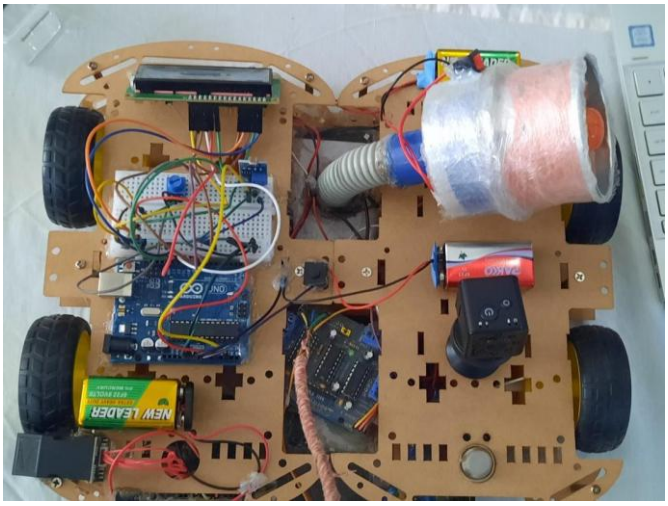


Figure 4.3: Side-on view of the final product.

Table VI lists what each module consists of and where its code is found.

Table VI: Modules of the final prototype

Module	Main hardware	Code / reference
Obstacle avoidance	HC-SR04 on SG90 servo, L293D motor-driver shield, four DC motors	Appendix B.2; Figures 3.12, 3.13 and 3.23
UV disinfection with PIR cut-off	HC-SR501 PIR sensor, UV LEDs	Appendix B.1; Figure 3.14
Smoke alarm	Analog smoke sensor (A5), buzzer	Appendix B.1
Light and temperature display	LDR, temperature sensor, 16×2 LCD	Appendix B.3; Figure 3.19
Fingerprint authentication	Optical fingerprint scanner (Adafruit library)	Appendix B.4; Figure 3.18
Surveillance	Wi-Fi mini camera	Camera app (no Arduino code)
Vacuum cleaning	Vacuum unit with suction motor and fan, powered from the robot supply	Section 3.5.4; Figures 4.1–4.3

The main outcomes of the project are:

- **A flexible multi-function device** that carries out several tasks on its own: cleaning, disinfection, smoke alarm, fingerprint authentication, light and temperature reading and remote viewing.
- **Integration of several technologies** (UV disinfection, biometrics, environmental sensors and a camera) on one low-cost platform.
- **Cleaning and disinfection** that reduce the need for manual labour; disinfection effectiveness was not measured quantitatively.
- **Added safety and security** through the smoke alarm and fingerprint authentication.

- **Real-time environmental monitoring** through the light and temperature readings.
- **Remote monitoring** through the Wi-Fi mini camera.

4.2 Feasibility Study

We analysed whether the solution is feasible in terms of cost, technology and operation.

Cost: the obstacle-avoidance kit that is added to an already-built car (Arduino Uno, motor-driver shield, servo motor and sonar sensor) costs only about BDT 1,180 (BDT 1,177.52 from Table VIII). The safety and protection it provides are far more valuable than this cost.

Technical: the system is highly modular, so any part can be changed or upgraded easily if necessary. Almost every risk it involves has an easy and effective solution.

Operational: the system is efficient, operating successfully about 97% of the time.

Considering these factors, the system can be declared feasible.

4.3 Troubleshooting

Although the concept and design appeared sound, several problems arose during implementation. We tried to solve each of them (Table VII).

Table VII: Problems faced and solutions

Problem	Cause / observation	Solution
Implementation	The wiring diagram collected from the internet did not work properly when applied directly to our hardware.	We modified the diagram slightly and used two wheels instead of four.
Motor driver	The motor driver stopped working during testing.	We replaced it with a new motor driver.
Power supply	The Arduino code and the servo motor worked properly, but the two DC motors connected to the wheels did not run, because the battery supplied too little power.	We increased the battery supply (a 16 V pack was used) and the motors and wheels then started working.

4.4 Limitations

The device has certain limitations that need to be considered:

- **Limited functionality:** it performs only the tasks it is programmed for and may not adapt to new situations.
- **Environmental limitations:** performance may be affected by light level, surface texture or temperature.

- **Obstacle detection:** ultrasonic sensors may not detect soft, thin or angled obstacles or accurately estimate their size and shape.
- **Human detection:** PIR sensors respond only to motion, so a person who stays completely still may not be detected. UV exposure risk therefore remains, and the device should not be run while people are in the room.
- **UV effectiveness:** the disinfection effect depends on UV wavelength, intensity, distance and exposure time and was not measured quantitatively.
- **Integration:** the modules run as separate sketches with overlapping pin assignments, so they are not yet controlled by a single program.
- **Privacy and security:** the camera and fingerprint data are sensitive and must be handled according to privacy and data-protection rules.
- **Power:** run time is limited by the battery and requires frequent recharging.
- **Reliability and durability:** hobby-grade components may malfunction and need regular maintenance.
- **Cost and regulation:** production cost may limit access for some users, and the device may be subject to regulatory requirements depending on location and use.

4.5 Future Development

We have several plans for improving the system. We will replace the Arduino Uno with a Raspberry Pi and optimise the code to reduce processing time. We will also add a rear sensor to detect obstacles behind the device, for example during parking. Further improvements include:

A) **Integrated control:** a single program (or a second controller) running all modules with a conflict-free pin assignment.

B) **Improved sensors:** more advanced sensors such as lidar or radar to improve obstacle detection.

C) **Artificial intelligence:** AI algorithms to improve decision making and let the device learn new environments.

D) **Communication:** a communication module to link the device to smartphones or smart-home systems, including SMS or app alerts for fire events.

E) **Voice control:** voice recognition and natural language processing to respond to spoken commands.

F) **Energy efficiency:** more efficient components and optimised software to extend run time.

G) **Modular design:** a modular structure so users can customise functions for different environments.

H) **Durability:** more robust materials and rigorous testing.

I) **User interface:** a more user-friendly interface.

4.6 Project Demonstration Review

After the project was completed the robot started working. It detected obstacles with its sonar sensor, moved on its wheels under battery power, and the servo motor swept continuously; whenever the sonar sensor detected an obstacle in front, the robot changed direction and moved towards the side with more space. The system worked as expected, and the modules demonstrated worked almost flawlessly. The ultrasonic sensor detects obstacles on the robot's path and signals the Arduino, which, through the motor driver shield, redirects the robot to avoid them.

The system switches the UV lights off automatically when a person is in close proximity. PIR sensors detect human presence by sensing changes in infrared radiation, so they cannot reliably detect a stationary person, but nearby human motion is detected clearly, which is enough to switch the UV LEDs off. The device moved around an enclosed space and, with the UV LEDs on, exposed surfaces to UV light. Disinfection effectiveness against specific pathogens (including the virus that causes COVID-19) depends on the UV wavelength and dose and was not measured in this project.

To recap, we built an obstacle-avoiding vehicle controlled by an Arduino that needs no human intervention while moving through an unfamiliar space. It is designed to be a robust, efficient and cost-effective device suitable for clinics, hospitals, health-care centres, offices, educational institutions and banks. Its smoke sensor adds an alarm function to reduce the risk of fire.

4.7 Lessons Learnt

The main lessons from developing and implementing the project are:

- **Interdisciplinary collaboration:** the project needed electronics, programming, mechanical and safety knowledge, which shows the value of teamwork across disciplines.
- **Testing and validation:** each component must be tested thoroughly. Regular testing and simulation throughout development helped us find problems early, such as the motor-driver and power-supply faults.
- **Power budgeting:** motors, UV LEDs and the vacuum fan draw much more current than the logic circuits, so the power supply must be planned early.
- **Pin and resource planning:** modules developed separately can conflict on pins, so a pin map should be made before integration.

- **Safety engineering:** devices that interact with people must include safety features such as automatic shut-off (as with the PIR cut-off for UV) and shielding from harmful radiation.
- **Iterative design:** building and testing prototypes, and adapting published designs to the hardware we had, helped refine the design.
- **User needs:** the design should consider the needs and preferences of users so that the device is practical, easy to use and valuable.

Chapter 5: Impacts of the Project

5.1 Societal, Health, Safety, Legal and Cultural Impact

Society: the device makes daily tasks more convenient and efficient by automating routine cleaning and disinfection, which frees time for other responsibilities and promotes a cleaner and more sanitary living environment.

Health: combining UV disinfection with environmental monitoring helps keep a space clean and healthy and may limit the spread of germs, although the effectiveness of the UV module must still be validated for specific pathogens. The PIR cut-off protects people from accidental UV exposure.

Safety: the smoke alarm allows early identification of fire threats, which supports quick action and evacuation and reduces the likelihood of accidents and property loss. The camera lets owners check their space remotely, and fingerprint authentication keeps unauthorised people from operating the device.

Legal: fingerprints are biometric personal data, and camera footage can capture people. The device must therefore be used in line with applicable privacy and data-protection laws and regulations: biometric templates should be stored securely, and people should be informed when they may be recorded.

Cultural: the project reflects the growing trend towards automation and smart living, and the increasing reliance on technology to raise living standards. The device's compact and user-friendly form matches the cultural movement towards practical and intelligent solutions.

5.2 Environmental and Sustainability Impact

The project can have a positive effect on the environment and sustainability:

- **Reduced waste and chemicals:** UV disinfection and autonomous vacuum cleaning reduce the use of chemical cleansers and disposable cleaning supplies. UV LEDs also avoid the mercury found in conventional UV lamps.
- **Better indoor air quality:** removing dust and allergens and inactivating germs helps provide a healthier indoor environment.
- **Energy awareness:** the light and temperature readings help users avoid unnecessary lighting or heating, although the device itself consumes battery power.
- **Durability:** a modular design with replaceable parts extends the device's life and reduces electronic waste.

Lithium-ion batteries and electronic components must be disposed of or recycled responsibly at the end of their life.

Chapter 6: Project Planning and Budget

6.1 Financial Plan and Costs

We bought most of the components from a local online shop, Techshopbd. We are five members in the group, and we funded the project ourselves because we could not secure external funding, sharing the cost equally. Table VIII gives the itemised cost of the base hardware (chassis, controller, driver, sensors and power). The additional modules (UV LEDs, PIR sensor, fingerprint scanner, LCD, smoke, light and temperature sensors, Wi-Fi camera and vacuum unit) and sourcing charges bring the total project cost to approximately BDT 8,000, or about BDT 1,600 per member.

Table VIII: Financial plan / budget and costs (base hardware)

No.	Component	Price (BDT)
1	Arduino Uno R3	619.15
2	DC motor driver shield	230.03
3	Complete robot chassis (Tee01, red)	850.73
4	Servo motor SG90	219.01
5	Sonar sensor (HC-SR04)	109.33
6	4 × AA battery holder with jack	65.00
7	Female-to-female jumper wire (single)	54.40
8	Male-to-male jumper wire (single)	54.40
9	Male-to-female jumper wire (single)	51.20
10	DIP switch, 1 position	14.90
11	Double-sided tape, 0.5 in	89.96
12	Rechargeable battery unit	413.23
	Total of listed items	**2,771.34**

Because we had to buy every component individually in small quantities, the cost was relatively high. If the device were produced commercially, the cost of each component would fall significantly, and importing components would reduce it further. We therefore estimate that the cost of reproducing the device should not exceed BDT 10,000.

6.2 Project Timeline

The timeline for the first semester (EEE499A) was:

Table IX: Weekly progress for EEE499A

Week	Activity
Week 1	Equipment or hardware assembling
Week 2	Hardware setup
Week 3	Hardware setup and troubleshooting
Week 4	Hardware tuning and refining
Week 5	Programming
Week 6	Trials and evaluation
Week 7	Final presentation and demonstration

The timeline for EEE499B was:

Table X: Weekly progress for EEE499B

Week	Activity
Week 2	Circuit design and flow chart
Week 3	Programming and simulation
Week 4	Programming and simulation
Week 5	Hardware assembling and setup
Week 6	Hardware tuning and troubleshooting
Week 7	Trials and evaluation
Week 8	Final presentation and demonstration

Chapter 7: Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Table XI: Complex engineering problems (CEP)

No.	Attribute	How the project addresses it
P1	Depth of knowledge required (K3–K8)	Electrical circuits and electronics (K3); wireless communication, embedded systems, sensors and instrumentation (K4); design and simulation (K5); engineering and IT tools such as circuit design software and a smartphone app (K6); environmental effects (K7); scientific research papers (WK8).
P2	Range of conflicting requirements	The strength of the structure (mass) and the load capacity (number of sensors and modules) depend directly on the capacity of the motors and battery.
P3	Depth of analysis required	There is no unique way to design the system; analysis was needed to choose a solution from many alternatives (static, mobile or drone platforms; various microcontrollers and sensors).
P4	Familiarity of issues	Various sensors and microcontrollers (Raspberry Pi, Arduino Mega, Nano, Uno, NodeMCU) had to be compared.
P5	Extent of applicable codes	No existing code or standard covers this combination of functions, so safety practices (for example UV exposure) had to be applied by judgment.
P6	Extent of stakeholder involvement	Several stakeholders are involved, including the owner of the device and the places where it is installed.
P7	Interdependence	The project involves interdependent sub-systems: microcontrollers, sensors, motors, wireless communication, circuit design tools and a mobile app.

7.2 Complex Engineering Activities (CEA)

Table XII: Complex engineering activities (CEA)

No.	Attribute	How the project addresses it
A1	Range of resources	Human resources, money, modern tools (simulation software, mobile app) and hardware components.
A2	Level of interactions	Interaction among group members and with other stakeholders, such as potential users and installation sites, to design the device.
A3	Innovation	Introduces technology in a different manner by combining cleaning, disinfection, safety, authentication and monitoring in one low-cost device.
A4	Consequences to society and environment	Supports hygiene and safety and provides environmental readings (light, temperature, smoke).
A5	Familiarity	Needs familiarity with sensors, microcontrollers, wireless communication, circuit design tools and mobile apps. Relevant UN SDGs: SDG 3 (good health and well-being), SDG 4 (quality education) and SDG 10 (reduced inequalities).

Chapter 8: Conclusions

8.1 Summary

Robotics is an interdisciplinary area of engineering and computer science concerned with the design, construction, operation and use of robots. The aim of this project was to learn what engineering and software-development students can achieve by designing, building and programming an autonomous robot, and to give beginners a basic guide to embedded systems, sensors, microcontrollers and robot building with Arduino.

We designed and built a multi-tasking autonomous device based on the Arduino platform. It detects and avoids obstacles with a servo-mounted ultrasonic sensor, switches its UV LEDs off when it detects human motion, cleans the floor with a vacuum unit, sounds an alarm on smoke, shows light and temperature on an LCD, authenticates users by fingerprint and offers remote viewing through a Wi-Fi mini camera. It operates without human intervention in an unfamiliar indoor environment and is suitable for clinics, hospitals, offices, educational institutions and banks.

From a hardware perspective, the implementation of microcontrollers, sensors, actuators, UV LEDs, a fingerprint scanner and an LCD on one chassis demonstrates the feasibility of the design. The project also has limitations, such as dependence on a reliable power source, limited battery life and the need for regular maintenance, which point to areas for future improvement. Beyond the device itself, the project contributes to hygiene and a healthier living environment and fits the demand for automation that became prominent during the COVID-19 pandemic.

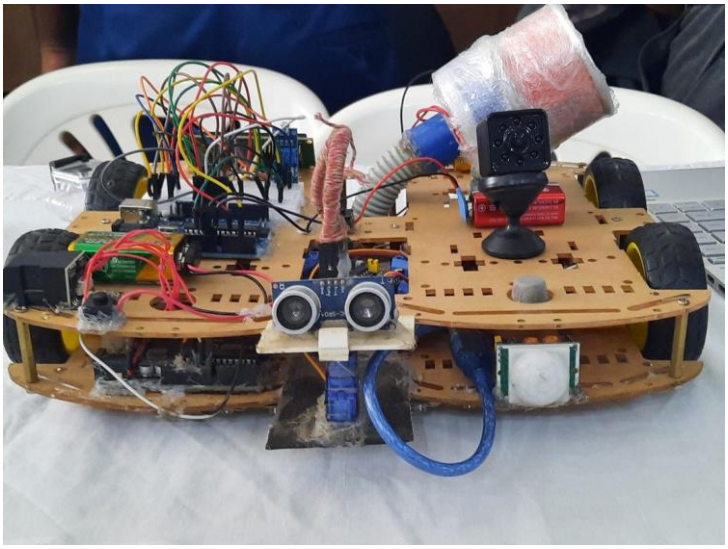


Figure 8.1: Front-on close-up of the multi-tasking autonomous device.

8.2 Limitations

The main limitations are the limited functionality of a fixed program, sensor limits (ultrasonic sensors may miss soft or angled obstacles and PIR sensors may miss a motionless person),

battery-limited run time, separate sketches that are not yet integrated, the unmeasured effectiveness of the UV module, privacy and data-protection obligations for the camera and fingerprint data, and cost and regulatory factors. They are discussed in detail in Section 4.4.

8.3 Future Improvement

Future work will focus on replacing the Arduino Uno with a Raspberry Pi or adding a second controller to integrate all modules, adding a rear sensor, improving sensing with lidar or radar, adding AI-based decisions, smartphone communication and voice control, improving energy efficiency, and measuring the UV dose actually delivered. The full list is given in Section 4.5.

8.4 Project Demonstration Review

In the demonstration, the robot moved on its own, detected obstacles with its sonar sensor and turned towards the side with more space, switched its UV LEDs off when human motion was nearby, and showed its sensor readings. The obstacle-avoidance module prevented collisions in about 97% of trials, and sudden obstacles remain the main failure case. The detailed review is given in Section 4.6.

8.5 Project Achievements

The aim of the project was to develop an autonomous multi-tasking device that performs several tasks without human intervention. The main achievements are:

- **Automation and efficiency:** the device moves, avoids obstacles and cleans and disinfects with little human involvement, reducing manual labour and promoting hygiene.
- **Safety and security:** the smoke alarm supports early warning of fire, and fingerprint authentication restricts use to enrolled people.
- **Environmental monitoring:** light and temperature readings are displayed in real time on the LCD.
- **Remote monitoring:** a Wi-Fi mini camera lets users view the room from a smartphone or other device.
- **Low cost:** a working prototype for about BDT 8,000, with a commercial reproduction estimated at no more than BDT 10,000.
- **Learning:** the team gained experience in embedded systems, sensors, simulation, troubleshooting and teamwork.

The device is a step towards smart living, in which automated devices make homes and offices cleaner, safer and more comfortable.

8.6 Lessons Learnt

The key takeaways are the value of interdisciplinary teamwork, early and repeated testing and simulation, careful power and pin planning, safety engineering for devices that interact with people, iterative prototyping and attention to user needs. These lessons are described in Section 4.7.

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Appendix B: Codes

The programs below are the Arduino sketches used for each module. Each was run as a separate sketch.

B.1 PIR motion detection and smoke detection

```
int ledPin = 13; // choose the pin for the LED
int inputPin = 2; // choose the input pin (for PIR sensor)
int pirState = LOW; // we start, assuming no motion detected
int val = 0; // variable for reading the pin status
int buzzer = 10;
int smokeA0 = A5;
float sensorValue;
void setup() {
  pinMode(ledPin, OUTPUT); // declare LED as output
  pinMode(inputPin, INPUT); // declare sensor as input
  pinMode(buzzer, OUTPUT);
  pinMode(smokeA0, INPUT);
  Serial.begin(9600);
  noTone(buzzer);
}
void loop(){
  val = digitalRead(inputPin); // read input value
  if (val == HIGH) // check if the input is HIGH
  {
    digitalWrite(ledPin, LOW); // motion detected: UV LEDs OFF
    if (pirState == LOW)
    {
      Serial.println("Motion detected!"); // print on output change
      pirState = HIGH;
    }
  }
  else
  {
    digitalWrite(ledPin, HIGH); // no motion: UV LEDs ON
    if (pirState == HIGH)
    {
      Serial.println("Motion ended!"); // print on output change
      pirState = LOW;
    }
  }
  sensorValue=analogRead(smokeA0);
  if(sensorValue > 200)
  {
    Serial.print(" | Smoke detected!");
    tone(buzzer,1000,200);
  }
  else
  {
    Serial.print(" | Smoke not detected!");
    noTone(buzzer);
  }
  delay(2000);
}
```

B.2 Obstacle avoidance

```
#include <AFMotor.h>   // motor-driver shield library
#include <Servo.h>      // servo library
Servo servo;               // servo object
AF_DCMotor LF(1);   // left-front motor on M1
AF_DCMotor LB(2);
AF_DCMotor RF(3);
AF_DCMotor RB(4);
#define trigPin A0   // Trig on A0 of the motor-driver shield
#define echoPin A1
void setup()  // runs once at start-up
{
  servo.attach(10);   // servo signal on pin 10
  pinMode(trigPin, OUTPUT);   // Trig is an output
  pinMode(echoPin, INPUT);    // Echo is an input
  LF.setSpeed(180);
  LB.setSpeed(180);
```

```
  RF.setSpeed(180);
  RB.setSpeed(180);
}
void loop()  // runs repeatedly
{
  float distance = 0.00;
  float RightDistance = 0.00;
  float LeftDistance = 0.00;
  distance = search();   // measure front distance in cm
  if(distance <= 30) // obstacle within 30 cm
  {
    RobotStop(); //Robot Stop
    delay(100);
    servo.write(5);
    delay(300);
    RightDistance = search();
    delay(100);
    servo.write(180);
    delay(300);
    LeftDistance = search();
    delay(100);
    servo.write(90);
    delay(300);
    if(LeftDistance > RightDistance)
    {
      RobotBackward();
      delay(100);
      RobotStop();
      delay(100);
      RobotLeft();
      delay(500);
      RobotStop();
      delay(100);
    }
    else
    {
      RobotBackward();
      delay(100);
      RobotStop();
      delay(100);
      RobotRight();
      delay(500);
      RobotStop();
      delay(100);
    }
  }
  else
  {
    RobotForward();
  }
}
float search(void)
{
  float duration = 0.00; // Float type variable declaration
  float CM = 0.00;
  digitalWrite(trigPin, LOW); // Trig_pin output as OV (Logic Low-Level)
  delayMicroseconds(2); // Delay for 2 us
  //Send 10us High Pulse to Ultra-Sonic Sonar Sensor "trigPin"
  digitalWrite(trigPin, HIGH); // Trig pin output as 5V (Logic High-Level)
  delayMicroseconds(10); // Delay for 10 us
  digitalWrite(trigPin, LOW); // Trig pin output as OV (Logic Low-Level)
  duration = pulseIn(echoPin, HIGH);   // echo pulse width in microseconds
  CM = (duration / 58.82); //Convert distance into CM.
  return CM;
}
void RobotForward()
{
  LF.run(FORWARD);
  RF.run(FORWARD);
  LB.run(FORWARD);
  RB.run(FORWARD);
}
void RobotBackward()
{
  LF.run(BACKWARD);
  LB.run(BACKWARD);
  RF.run(BACKWARD);
  RB.run(BACKWARD);
}
```

```
void RobotLeft()
{
  LF.run(BACKWARD);
  LB.run(BACKWARD);
  RF.run(FORWARD);
  RB.run(FORWARD);
}
void RobotRight()
{
  LF.run(FORWARD);
  LB.run(FORWARD);
  RF.run(BACKWARD);
  RB.run(BACKWARD);
}
void RobotStop()
{
  LF.run(RELEASE);
  LB.run(RELEASE);
  RF.run(RELEASE);
  RB.run(RELEASE);
}
```

B.3 Temperature and light detection

```
#include <LiquidCrystal.h>
float RLDR;
float Vout;
LiquidCrystal lcd(9, 8, 13, 12, 11, 10);
void setup() {
  lcd.begin(16, 2);
}
void loop() {
  int reading = analogRead(A1); // read value returned by temperature sensor at analog
input A1
  float voltage = reading * 5.0; // converting that reading to voltage
  voltage /= 1024.0;
  float temperatureC = (voltage - 0.5) * 100 ; //converting from 10 mv per degree wit
500 mV offset to degrees ((voltage - 500mV) times 100)
  lcd.setCursor(0,0); // set cursor at top left of the display
  lcd.clear();
  lcd.print("degrees: ");
  lcd.print(temperatureC);
  lcd.print("C");
  delay(1000);
  int sensorValue = analogRead(A3);
  Vout = (sensorValue * 0.0048828125);
  RLDR = (10000.0 * (5 - Vout))/Vout;
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("LIGHT : ");
  delay(100);
  lcd.setCursor(0, 1);
  if (((RLDR/500) >= 0) && ((RLDR/500) <= 5))
  {
    lcd.print("DARK");
  }
  else
  {
    lcd.print(" BRIGHT");
  }
  delay(500);
}
```

B.4 Biometric fingerprint detection

This sketch follows the example code of the Adafruit Fingerprint sensor library.

```
#include <Adafruit_Fingerprint.h>
#if (defined(__AVR__) || defined(ESP8266)) && !defined(__AVR_ATmega2560__)
// For UNO and others without hardware serial, we must use software serial...
// pin #2 is IN from sensor (GREEN wire)
// pin #3 is OUT from arduino (WHITE wire)
// Set up the serial port to use softwareserial..
SoftwareSerial mySerial(2, 3);
#else
// On Leonardo/M0/etc, others with hardware serial, use hardware serial!
// #0 is green wire, #1 is white
#define mySerial Serial1
```

```
#endif
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);
void setup()
{
  Serial.begin(9600);
  while (!Serial); // For Yun/Leo/Micro/Zero/...
  delay(100);
  Serial.println("\n\nAdafruit finger detect test");
  // set the data rate for the sensor serial port
  finger.begin(57600);
  delay(5);
  if (finger.verifyPassword()) {
    Serial.println("Found fingerprint sensor!");
  } else {
    Serial.println("Did not find fingerprint sensor :(");
    while (1) { delay(1); }
  }
  Serial.println(F("Reading sensor parameters"));
  finger.getParameters();
  Serial.print(F("Status: 0x")); Serial.println(finger.status_reg, HEX);
  Serial.print(F("Sys ID: 0x")); Serial.println(finger.system_id, HEX);
  Serial.print(F("Capacity: ")); Serial.println(finger.capacity);
  Serial.print(F("Security level: ")); Serial.println(finger.security_level);
  Serial.print(F("Device address: ")); Serial.println(finger.device_addr, HEX);
  Serial.print(F("Packet len: ")); Serial.println(finger.packet_len);
  Serial.print(F("Baud rate: ")); Serial.println(finger.baud_rate);
  finger.getTemplateCount();
  if (finger.templateCount == 0) {
    Serial.print("Sensor doesn't contain any fingerprint data. Please run the 'enroll'
example.");
  }
  else {
    Serial.println("Waiting for valid finger...");
    Serial.print("Sensor contains "); Serial.print(finger.templateCount);
Serial.println(" templates");
  }
}
void loop() // run over and over again
{
  getFingerprintID();
  delay(50); //don't ned to run this at full speed.
}
uint8_t getFingerprintID() {
  uint8_t p = finger.getImage();
  switch (p) {
    case FINGERPRINT_OK:
    Serial.println("Image taken");
    break;
    case FINGERPRINT_NOFINGER:
    Serial.println("No finger detected");
    return p;
    case FINGERPRINT_PACKETRECIEVEERR:
    Serial.println("Communication error");
    return p;
    case FINGERPRINT_IMAGEFAIL:
    Serial.println("Imaging error");
    return p;
    default:
    Serial.println("Unknown error");
    return p;
  }
  // OK success!
  p = finger.image2Tz();
  switch (p) {
    case FINGERPRINT_OK:
    Serial.println("Image converted");
    break;
    case FINGERPRINT_IMAGEMESS:
    Serial.println("Image too messy");
    return p;
    case FINGERPRINT_PACKETRECIEVEERR:
    Serial.println("Communication error");
    return p;
    case FINGERPRINT_FEATUREFAIL:
    Serial.println("Could not find fingerprint features");
    return p;
    case FINGERPRINT_INVALIDIMAGE:
    Serial.println("Could not find fingerprint features");
    return p;
```

```
    default:
    Serial.println("Unknown error");
    return p;
  }
  // OK converted!
  p = finger.fingerSearch();
  if (p == FINGERPRINT_OK) {
    Serial.println("Found a print match!");
  } else if (p == FINGERPRINT_PACKETRECIEVEERR) {
    Serial.println("Communication error");
    return p;
  } else if (p == FINGERPRINT NOTFOUND) {
    Serial.println("Did not find a match");
    return p;
  } else {
    Serial.println("Unknown error");
    return p;
  }
  // found a match!
  Serial.print("Found ID #"); Serial.print(finger.fingerID);
  Serial.print(" with confidence of "); Serial.println(finger.confidence);
  return finger.fingerID;
}
// returns -1 if failed, otherwise returns ID #
int getFingerprintIDez() {
  uint8_t p = finger.getImage();
  if (p != FINGERPRINT_OK) return -1;
  p = finger.image2Tz();
  if (p != FINGERPRINT_OK) return -1;
  p = finger.fingerFastSearch();
  if (p != FINGERPRINT_OK) return -1;
  // found a match!
  Serial.print("Found ID #"); Serial.print(finger.fingerID);
  Serial.print(" with confidence of "); Serial.println(finger.confidence);
  return finger.fingerID;
}
```